

LEONORA AP, Australia

WMO: 944500

Lat: 28.879S

Lon: 121.319E

Elev: 1217

StdP: 14.06

Time Zone: 8.00 (E08)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.5	39.9	12.9	11.1	83.7	15.9	12.8	79.3	22.7	66.2	20.8	64.3	2.3	110	0.491

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.9	107.3	64.9	104.5	64.5	101.5	64.3	72.9	85.4	71.2	86.1	69.6	86.0	11.5	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.6	113.9	78.9	67.3	104.9	76.9	64.7	95.8	75.7	37.5	85.5	35.9	85.8	34.4	86.3	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.5	19.2	17.6	32.6	111.7	1.5	2.6	31.5	113.5	30.6	115.0	29.8	116.5	28.7	118.4		
(5)			30.1	75.1	1.9	2.0	28.7	76.5	27.6	77.7	26.5	78.8	25.2	80.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.1	85.5	82.6	77.3	71.1	61.3	55.7	54.4	57.9	63.3	72.8	77.4	82.4
	DBStd	12.64	7.60	7.96	7.28	6.29	5.56	4.99	4.98	5.98	6.80	7.28	7.37	7.32
	HDD50	24	0	0	0	0	0	6	14	4	0	0	0	0
	HDD65	1138	1	1	3	13	142	281	330	232	114	17	4	0
	CDD50	7353	1100	912	848	632	351	178	150	250	400	706	821	1005
	CDD65	2994	636	493	385	196	29	3	2	13	64	258	375	540
	CDH74	39313	9297	6687	4494	1932	304	19	27	201	912	3176	4765	7499
	CDH80	21985	5899	4026	2301	769	62	0	1	31	323	1532	2572	4469
(14) Wind	WSAvg	9.2	11.5	10.7	10.0	8.1	7.3	7.4	7.1	8.2	9.0	10.1	10.7	10.9
(15) Precipitation	PrecAvg	9.8	0.7	1.7	1.1	0.8	1.1	1.0	0.9	0.6	0.4	0.5	0.6	0.7
	PrecMax	21.7	5.0	11.2	7.0	4.0	4.3	3.9	4.0	3.1	1.9	2.2	2.0	3.7
	PrecMin	3.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	4.3	1.1	2.3	1.5	0.9	1.1	0.9	1.0	0.7	0.4	0.6	0.5	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	112.0	109.7	103.8	97.5	85.6	76.5	77.5	84.3	92.5	99.8	106.5	108.0
		MCWB	66.1	64.9	63.6	63.1	57.8	55.9	55.1	54.7	57.4	60.8	62.5	64.5
	2%	DB	107.9	105.8	99.4	92.1	81.0	72.6	72.9	79.5	87.8	96.6	101.2	105.5
		MCWB	65.6	64.8	64.3	62.0	56.4	54.1	53.7	54.1	56.6	60.1	61.8	63.8
	5%	DB	104.9	102.3	95.8	88.2	77.1	69.4	69.2	75.5	83.8	92.7	97.2	102.0
		MCWB	65.7	65.0	63.6	61.3	56.3	53.0	52.3	53.0	55.6	59.2	61.0	63.6
	10%	DB	101.3	98.2	91.8	84.3	73.7	66.4	66.2	71.6	79.3	89.0	92.9	98.0
		MCWB	65.3	65.5	63.7	60.4	55.6	52.0	51.1	52.6	54.7	58.3	60.4	63.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.4	75.4	71.2	68.4	63.4	59.9	61.1	59.7	60.8	65.5	67.8	72.3
		MCDB	86.8	88.2	82.7	82.6	73.4	68.4	70.2	70.7	79.3	82.6	83.2	88.3
	2%	WB	73.3	73.3	69.2	65.7	60.6	57.9	56.8	57.5	58.7	63.2	66.1	69.9
		MCDB	84.5	85.3	82.9	81.0	73.0	66.4	65.7	69.2	79.6	82.4	83.5	87.1
	5%	WB	71.3	71.5	68.0	64.2	58.8	56.1	55.0	55.7	57.2	61.7	64.5	67.8
		MCDB	88.1	84.6	82.7	79.0	71.6	64.8	64.0	69.5	78.6	83.2	85.5	87.9
	10%	WB	69.6	69.8	66.6	62.9	57.0	54.3	53.2	53.9	55.7	60.2	63.1	65.7
		MCDB	88.6	84.7	82.3	77.3	70.2	62.6	63.0	68.3	76.5	84.0	85.2	90.0
(35) Mean Daily Temperature Range	5% DB	MDBR	22.9	21.6	20.4	21.8	22.5	20.2	22.0	23.9	25.2	24.6	23.8	23.3
		MCDBR	27.1	26.5	25.6	26.1	26.5	24.1	27.0	28.9	31.8	29.5	28.6	26.5
		MCWBR	7.0	7.2	7.5	8.0	10.6	11.2	12.9	12.4	12.5	10.3	9.2	8.1
	5% WB	MCDBR	22.0	20.4	19.4	19.8	20.7	18.0	19.9	22.3	25.9	24.6	24.1	23.9
		MCWBR	7.2	7.1	6.9	7.7	8.9	9.6	10.9	10.7	11.1	9.4	7.9	8.1
(40) Clear-Sky Solar Irradiance	taub	0.342	0.345	0.325	0.312	0.289	0.278	0.276	0.283	0.301	0.321	0.328	0.331	
	taud	2.579	2.590	2.621	2.629	2.653	2.665	2.660	2.618	2.556	2.513	2.529	2.567	
	Ebn at Noon	317	311	307	297	290	287	292	302	310	314	319	321	
	Edh at Noon	34	32	30	27	23	22	23	26	31	35	35	34	
(44) All-Sky Solar Radiation	RadAvg	2473	2199	1906	1546	1244	1049	1158	1494	1929	2244	2482	2599	
	RadStd	196	182	136	82	80	67	77	81	78	139	107	144	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page